

DIFFERENTIAL DIAGNOSIS

Thrombocytopenia may result in purpura; however, these lesions are flat. Other causes of non-palpable cutaneous purpura include local trauma to the skin of aged individuals with vascular fragility, chronic sun exposure, and endogenous or iatrogenic hypercorticism. Erythema elevatum diutinum may resemble Kaposi sarcoma or multicentric reticulohistiocytosis. Scurvy may present as hemorrhagic, follicular papules over the lower extremities. Various infectious agents may be associated with hemorrhagic skin lesions due to disseminated intravascular coagulation or septic emboli.

The cutaneous manifestations of disseminated intravascular coagulation, known as purpura fulminans, appear as extensive areas of purpura with a slate-gray color. These lesions occur in patients with acute meningococcemia but also with gram-positive coccal sepsis; skin biopsy specimens show thrombosis but not necrotizing venulitis. Septic emboli present as a limited number of hemorrhagic pustules, papules, and vesicles distributed over acral areas. Rarely, echo and coxsackievirus infections may cause purpuric papules. Rocky Mountain spotted fever typically begins as purpuric papules over the ankles and wrists that spread to involve the trunk. Cholesterol embolization syndrome includes purpura, livedo reticularis, ulcers, and nodules.

The progressive pigmentary purpuric dermatoses (purpura simplex), which encompass several disorders, appear as macular, petechial, pigmented, or lichenoid lesions.

The differential diagnosis of cutaneous nodules includes erythema nodosum, panniculitis, superficial forms of thrombophlebitis, fat necrosis associated with pancreatic disease, and systemic vasculitides such as polyarteritis nodosa, granulomatosis with polyangiitis (formerly Wegener granulomatosis), and eosinophilic granulomatosis with polyangiitis (Churg-Strauss syndrome).

TREATMENT

Therapeutic approaches may be divided into removal of the antigen, treatment of an underlying disease, and direct treatment of cutaneous necrotizing venulitis (CNV). Management of necrotizing vasculitis involves prevention of immune complex deposition, suppression of the inflammatory response, modulation of underlying immunopathologic mechanisms, and local therapy. When the eruption is associated with a precipitating event, withdrawal of the medication or treatment of the underlying infection results in resolution of the cutaneous lesions. If a coexisting chronic disease is present, treating the underlying condition may improve the cutaneous vascular manifestations.

In many cases, CNV is self-limited, and double-blind, placebo-controlled, prospective trials of therapeutic agents are generally lacking.

The treatment of CNV can be divided into two phases (Table 164-4), each based on an assessment of cutaneous disability as well as the toxicity and side effects of the therapeutic agents.

TABLE 164-4 Algorithm for Treatment of Cutaneous Necrotizing Venulitis

Phase One

- **H₁ antihistamines** – used to alleviate symptoms of palpable purpura and potentially reduce tissue deposition of circulating immune complexes
- **Nonsteroidal anti-inflammatory agents** – often combined with H₁ antihistamines
- **Colchicine**
- **Hydroxychloroquine sulfate**
- **Dapsone**

Phase Two

- Systemic glucocorticoids
- Azathioprine
- Cyclophosphamide
- Intravenous immunoglobulin (IVIG)
- Cyclosporine
- Plasmapheresis